Hunter Mills

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Santa Rosa, CA

Seasoned Data Scientist and Machine Learning Engineer with 8+ years of end-to-end AI product delivery. Proven ability to translate challenges into scalable, production-grade solutions that integrate with medical systems, cloud platforms, and embedded applications. 15+ publications ranging from rare disease prediction, bias mitigation, to deep learning on EKG.

Experience

Senior Machine Learning Engineer — October 2025 - Present
Visual Thought Labs — San Francisco, CA
- Developed ambient drug OCR in VA operating rooms, deployed on Nvidia Jetson at 77% lower hardware cost and 73% lower latency; built the SQL pipeline populating it, and HL7 system issuing results.

Senior Data Scientist — January 2022 - October 2025
UCSF — San Francisco, CA
- Built ”Drifter,” a data-drift monitoring tool used by 80 researchers across six hospitals; surfaced feature, label, and prediction drift that triggered timely ETL updates.
- Directed a nation-wide matched-cohort study on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR = 1.67) and publishing findings in JAMA Network Open.

Machine Learning Engineer — October 2021 - January 2022
Syntegra — San Francisco, CA
- Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.

Lead Data Scientist — December 2020 - October 2021
LifeBell AI — Atlanta, GA (remote)
- Deployed early sepsis detection system using XGBoost, with clinical explainer at Naval Medical Center San Diego.
- Designed a Dynamic-Time-Warped Gaussian Mixture Model for phenotype discovery, surfacing sepsis-risk subgroups several hours before conventional detection.

Data Scientist — August 2018 - December 2020
UCSF — San Francisco, CA
- Developed deep-learning EKG classifier for pulmonary hypertension, achieving an AUC of 0.89 and early-diagnosis capability up to two years before conventional testing.
- Led feasibility analyses that helped secure $5M+ in research partnerships (J&J, Roche, Genentech).

Embedded Systems Engineer — May 2014 - August 2018
NASA Education & Public Outreach — Rohnert Park, CA
- Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.
- Developed a STEM curriculum (“Learning By Making”) that raised high-school science scores by 7.8% versus controls.

Education

Stanford University	MS in Computational and Mathematical Engineering	2018
Sonoma State University	BS in Physics and Mathematics	2014
Santa Rosa Junior College	AS in Mathematics	2011

Skills

Languages	Python, SQL, Julia, C, C++, JavaScript, TypeScript
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, XGBoost, CVXPY, RAPIDS, JAX, TensorRT, Ollama, HuggingFace, StatsModels, lifelines
Infrastructure & Compute	Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker

Publications

- T. D. Brender, N. Stumpf, K. Vossler, S. Kim, **H. Mills**, A. Lee, H. Y. Yang, T. Heintz, E. Espejo, J. Boscardin, et al. "How ICU clinicians document 'Futility': A 10-year analysis of critical care notes using natural language processing notes". In: *Journal of Critical Care* 94 (2026), p. 155526.
- C. W. Cu, T. Heintz, N. Dundas, H. Farhan, **H. Mills**, J. W. Lee, C. Williams, A. W. Wallace, and J. Cobert. "Time and Motion Analysis of Controlled Substance Disposals: A Study of Workflows at a Single Center using Automated Dispensing Cabinets". In: *Anesthesiology Open* 1.1 (2026), e0006.
- N. E. Dundas, T. Law, T. Brender, **H. Mills**, E. Espejo, T. A Heintz, A. W. Wallace, and J. Cobert. "All That Shines Is Not Gold: Maintaining Scientific Rigor When Evaluating, Interpreting, and Reviewing Studies Using Large Language Models". In: *Anesthesiology* 144.2 (2026), pp. 272–288.
- E. Farrand, A. Chung, J. Joshua, H. Dong, **H. Mills**, A. Lee, M. Ieong, L. Radhakrishnan, O. Gologorskaya, and A. Butte. "Development and validation of a generalisable machine learning algorithm for identifying interstitial lung disease cohorts: a retrospective cohort study". In: *Eclinicalmedicine* 93 (2026).
- C. Halabi, **H. Mills**, A. M. DiGiorgio, G. Schenk, S. Israni, C. Peek-Asa, and G. T. Manley. "Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses". In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- C. W. Cu, N. E. Dundas, T. Heintz, Z. A. Sheikh, B. Alonso-Bermudez, J. Walker, A. Wooten, A. Badathala, A. Chapman, O. Ehie, K. Raghunathan, **H. Mills**, et al. "Validity of two subjective skin tone scales and its implications on healthcare model fairness". In: *NPJ Digital Medicine* 8.1 (2025), p. 595.
- S. Marini, A. R. Alwakeal, **H. Mills**, J. D. Bernstock, A. Mashlah, M. T. Hassan, J. Gerstl, F. Radmanesh, G. Schenk, S. Israni, et al. "Traumatic brain injury and risk of malignant brain tumors in civilian populations". In: *JAMA Network Open* 8.8 (2025), e2528850.
- G. Marquez-Grap, A. Leung, F. Orcales, A. Kranyak, C. Johnson, P. Smith, K. Haran, **H. Mills**, G. Schenk, A. Kahlon, et al. "Monitoring atopic dermatitis using mobile-app based photography and surveys". In: *Journal of Dermatological treatment* 36.1 (2025), p. 2555983.
- A. Mashlah, S. Marini, **H. Mills**, T. Yahya, F. Radmanesh, J. Chalif, B. E. Zusman, J. D. Bernstock, M. H. Al Mansi, R. Grashow, et al. "Traumatic Spinal Cord Injury and Subsequent Risk of Developing Chronic Cardiovascular, Neurologic, Psychiatric, and Endocrine Disorders". In: *JAMA Network Open* 8.11 (2025), e2541157.
- S. K. McGowan, M.-J. Corrales-Martinez, T. Brender, A. K. Smith, S. Kim, K. L. Harrison, **H. Mills**, A. Lee, D. Bamman, and J. Cobert. "Unclear Trajectory and Uncertain Benefit: Creating a Lexicon for Clinical Uncertainty in Patients with Critical or Advanced Illness Using a Delphi Consensus Process". In: *Medical Decision Making* 45.1 (2025), pp. 34–44.
- S. Miramontes, U. Khan, S. Zimmerman, E. L. Ferguson, **H. Mills**, B. Oskotsky, E. Phelps, T. T. Oskotsky, A. J. Capra, M. Glymour, et al. "Diagnostic Reversion in Dementia Care: A Real-World Analysis of Mild Cognitive Impairment Diagnoses Following Dementia in a Large Electronic Medical Record System". In: *medRxiv* (2025).
- H. Y. Yang, K. Raghunathan, E. Widera, S. Z. Pantilat, T. Brender, T. A. Heintz, E. Espejo, J. Boscardin, **H. Mills**, A. Lee, et al. "Lexical associations can characterize clinical documentation trends related to palliative care and metastatic cancer". In: *Scientific Reports* 15.1 (2025), p. 17245.
- J. Cobert, E. Espejo, J. Boscardin, **H. Mills**, D. Ashana, K. Raghunathan, T. A. Heintz, A. C. Chapman, A. K. Smith, and S. Lee. "Variation in Mentions of Race and Ethnicity in Notes in Intensive Care Units Across a Health Care System". In: *American Journal of Critical Care* 33.6 (2024), pp. 462–466.
- J. Cobert, **H. Mills**, A. Lee, O. Gologorskaya, E. Espejo, S. Y. Jeon, W. J. Boscardin, T. A. Heintz, C. J. Kennedy, D. C. Ashana, et al. "Measuring implicit bias in ICU notes using word-embedding neural network models". In: *Chest* 165.6 (2024), pp. 1481–1490.
- C. Halabi, S. Izzy, A. DiGiorgio, **H. Mills**, F. Radmanesh, J. Yue, et al. *Traumatic Brain Injury and Risk of Incident Comorbidities. JAMA Netw Open 7: e2450499–e2450499*. 2024.
- S. Kim, **H. Mills**, T. Brender, S. McGowan, E. Widera, A. C. Chapman, K. L. Harrison, S. Lee, A. K. Smith, D. Bamman, et al. ""My Mom Is a Fighter": A Qualitative Analysis of the Use of Combat Metaphors in ICU Clinician Notes". In: *Chest* 166.5 (2024), pp. 1162–1172.

- A. Tang, K. Rankin, G. Cerono, S. Miramontes, **H. Mills**, J. Roger, B. Zeng, C. Nelson, K. Soman, S. Woldemariam, et al. *Leveraging electronic health records and knowledge networks for Alzheimer's disease prediction and sex-specific biological insights. Nat Aging 4, 379–395*. 2024.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, L. Klein, N. Mantri, B. Rubin, J. Barrios, C. Chehoud, E. Kogan, et al. "Electrocardiogram detection of pulmonary hypertension using deep learning". In: *Journal of cardiac failure* 29.7 (2023), pp. 1017–1028.
- E. Farrand, O. Gologorskaya, **H. Mills**, L. Radhakrishnan, H. R. Collard, and A. J. Butte. "Machine-learning algorithm to improve cohort identification in interstitial lung disease". In: *American Journal of Respiratory and Critical Care Medicine* 207.10 (2023), pp. 1398–1401.
- M. Mahendra, Y. Luo, **H. Mills**, G. Schenk, A. J. Butte, and R. A. Dudley. "Impact of different approaches to preparing notes for analysis with natural language processing on the performance of prediction models in intensive care". In: *Critical care explorations* 3.6 (2021), e0450.